

Satheesh's Terrace Farm Setup Manual

Step-by-step beginner guide from empty terrace to protected smart organic farm

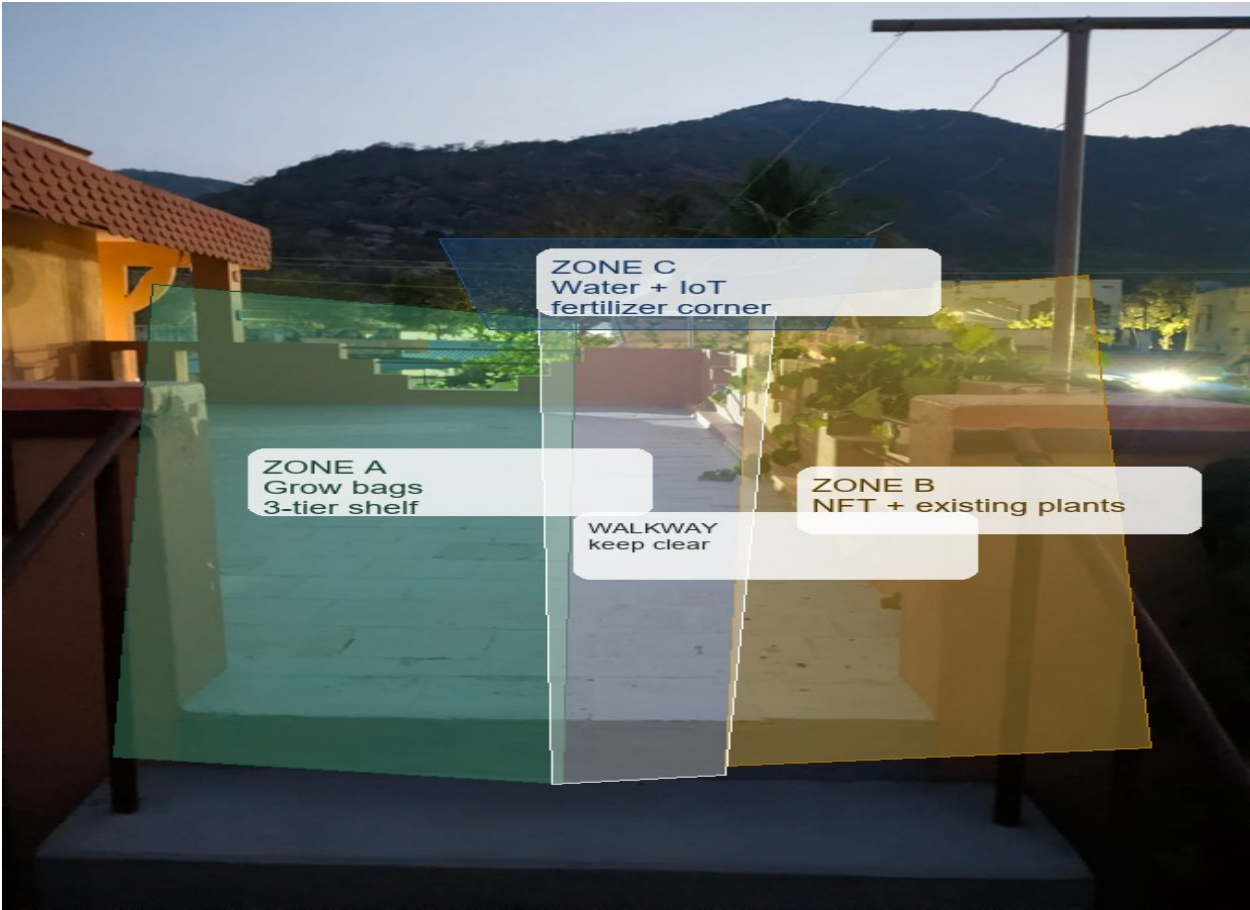
Based on the terrace photo provided: strong parapet walls, open walking space, existing right-side plants, hill-side humidity, and leakage risk.



Your terrace photo used for this setup plan.

Golden rule: protect the terrace first, then build the farm. A farm that leaks is not a farm; it becomes a building problem.

1. What I See On Your Terrace



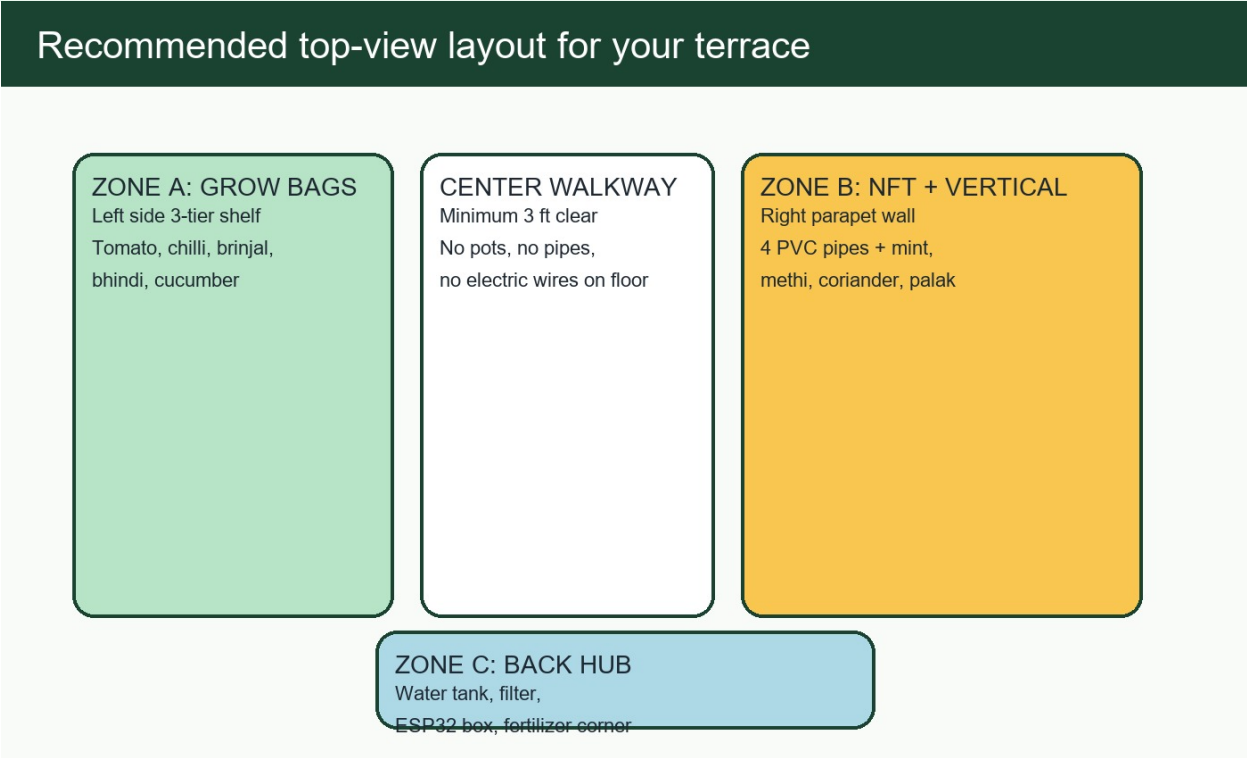
Recommended zones overlaid on your terrace photo.

Observation from photo	What it means for setup
Long clear rectangular terrace	Excellent for left/right zoning with a center walkway.
Strong parapet walls	Good for NFT mounting, trellis, cameras, shade-net hooks and pipe support.
Existing plants on right side	Right side already behaves like a green wall; expand it into NFT + vertical garden.
Open hill side and sky	Good airflow and humidity, useful for mint, leafy greens and creepers.
Tile floor with leakage concern	All systems must be raised with trays and drainage; no direct wet load on tiles.
Overhead metal/pipe structure visible	Can help support shade net, but check strength before hanging load.

My verdict: your terrace is suitable, but the design must be clean, raised, and modular. Do not cover the whole floor with pots. Keep the center open so you can inspect, harvest, clean, and repair leaks.

2. Final Layout For Your Terrace

This layout is chosen to match the actual photo, not a generic terrace drawing.



Top-view layout: left grow bags, center walkway, right NFT, back infrastructure hub.

Zone	Position	Use	Why this is best
Zone A	Left side from entry	3-tier grow bag shelf	Open space, easy access, less disturbance to existing plants.
Zone B	Right parapet	NFT pipes + existing plants + vertical garden	Already has greenery and wall support; pipes can mount above floor.
Zone C	Back/hill side corner	Water tank, filter, ESP32, fertilizer corner	Close to parapet, easy to route rainwater and wiring.
Center	Middle walkway	3 ft clear path	Maintenance, harvesting, emergency repair and leak inspection.

3. Build Order From Scratch

Do not start by buying plants. Start by making the terrace ready. This is how experienced farmers avoid expensive rework.

Phase	Build work	Result
0	Measure, clean, inspect leaks, decide drain direction	You know exact working area and weak spots.
1	Waterproof protection and raised base system	Terrace safe from daily water.
2	Install left grow-bag shelf and right NFT wall clamps	Farm structure ready without plants.
3	Install water storage, reservoir, gutters and drainage trays	Water movement controlled.
4	Add grow bags, soil mix, trellis and NFT pipes	Crop zones ready.
5	Install sensors, cameras, ESP32 box and electrical safety	Monitoring ready.
6	Plant only 30-40% capacity first, then expand	Less risk while you learn.

Old farmer lesson: first season is not for showing off. First season is for learning your terrace's water, heat, wind and pest behavior. After that, expansion becomes easy.

4. Phase 0: Measure And Mark

Beginner work: one tape, one chalk, one notebook. Do this before buying material.

Task	How to do it	Why
Measure length and width	Use tape; write exact feet/inches.	Your shelf, pipe and tank decisions depend on real size.
Mark 3 ft walkway	Chalk a center walking lane from entry to back.	This must remain empty forever.
Find drain direction	Pour 1 bucket water and watch flow.	All tray outlets should point toward drain.
Leak inspection	Check floor cracks, parapet corners, tile joints.	Repair before placing farm weight.
Sun observation	Take photos at 8am, 12pm, 4pm.	Decides crop positions and shade net.
Wind observation	Watch which side gets strong wind.	Trellis and tall plants need support there.

5. Phase 1: Terrace Protection

Because you already have leakage concern, this is the most important chapter.

Terrace protection stack before farming

Plants / grow bags / NFT reservoir

Raised stand or plastic pallet: 2-4 inch air gap

Drain tray + overflow pipe to terrace drain

Geotextile or protection sheet

Waterproof membrane / coating

Existing terrace slab and tiles

Protection stack: farming system sits above the terrace, not directly on the terrace.

- Repair cracks and parapet joints before farm installation.
- Add waterproof membrane/top coat if leakage continues after damp-proof paint.
- Place geotextile/protection sheet under active farm zones.
- Use plastic pallets, PVC stands or GI stands to create 2-4 inch air gap.
- Every grow bag shelf must have a drain tray or gutter below.
- Every reservoir or fertilizer drum must sit inside a catch tray.
- Keep drain outlet open and add leaf mesh so compost/leaves do not block it.

Do not trust only paint. Paint protects surface, but daily farming creates continuous wetting. Your protection must include drainage and air gap.

6. Phase 2: Build Zone A Grow Bag Shelf

This side will grow fruiting vegetables. Fruiting crops need root depth, support and stable moisture.

3-tier grow bag shelf arrangement

Top shelf
Bhindi, beans, herbs
lighter bags

Middle shelf
Chilli, capsicum,
cucumber with trellis

Bottom shelf
Tomato, brinjal
heaviest bags

Floor must stay dry
Use tray + air gap

3-tier shelf plan for left side.

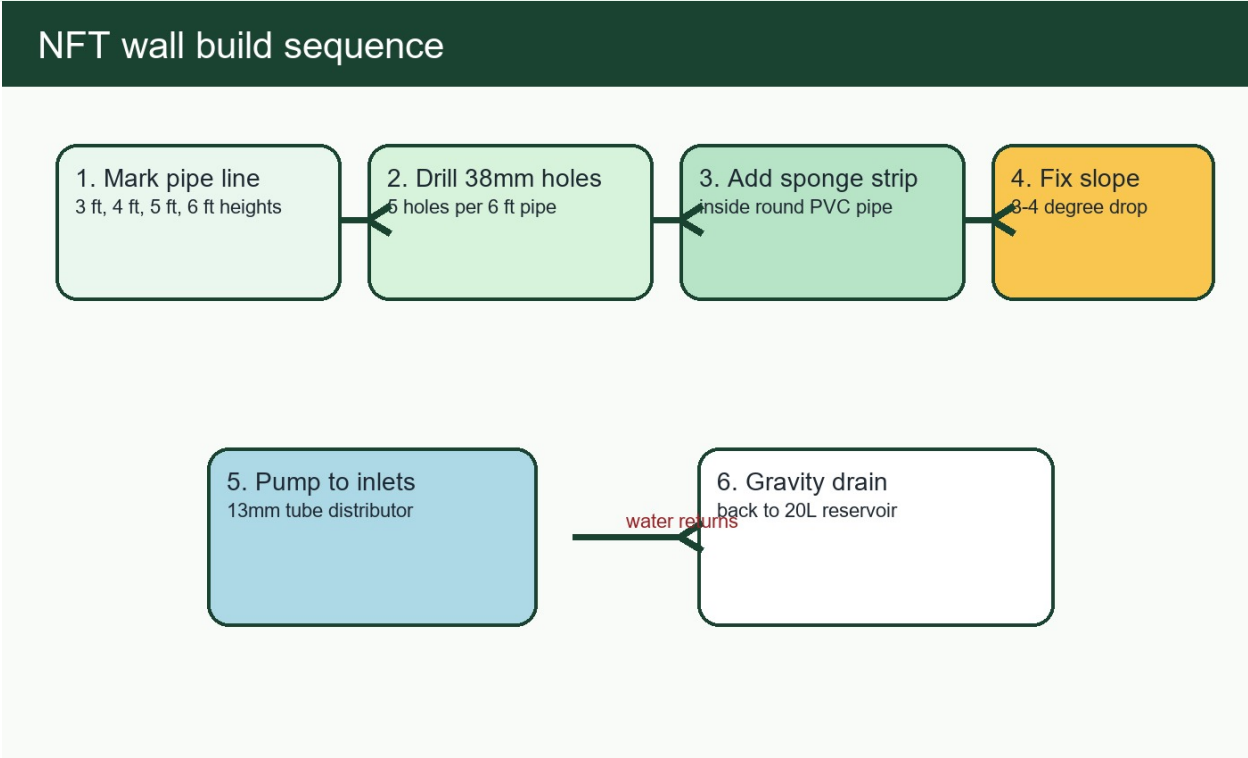
Part	Specification	Beginner note
Frame	1 inch GI square pipe or treated bamboo	GI is stronger; bamboo cheaper but needs replacement.
Size	About 6 ft wide x 10-12 ft long x 6 ft high	Keep inside terrace wall line.
Shelves	3 levels: 2 ft, 4 ft, 6 ft	Do not overload top shelf with heavy wet bags.
Bag placement	Heaviest bags bottom	Tomato/brinjal below; herbs/beans above.
Drainage	Tray/gutter below each shelf	Water must return to drain, not floor.
Support	Tie frame to parapet/wall if windy	Prevents falling during storms.

Recommended first crop load

Crop	Bags first month	Bag size	Why
Tomato	2	24x24 inch	Learn staking and watering first.
Chilli	3	15x15 inch	Good for Chittoor heat.
Brinjal	2	15x15 inch	Strong terrace crop.
Bhindi	4	12x12 inch	Beginner-friendly summer crop.
Cucumber	1	15x15 inch + trellis	Needs moisture and training.
Herbs	4-6 small pots	10x10 inch	Fast confidence crops.

7. Phase 3: Build Zone B NFT Wall

Right side is best for NFT because the parapet can carry pipe clamps and wiring can run along wall.



NFT build sequence for right parapet wall.

Step	Action	Exact practical instruction
1	Cut PVC pipes	Use 2.5 inch round PVC, 6 ft each, 4 pipes.
2	Mark net pot holes	5 holes per pipe, about 15cm spacing; keep ends clear.
3	Drill holes	Use 38mm hole saw; drill slowly to avoid cracking.
4	Add foam strip	55mm wide kitchen sponge strip along pipe bottom to spread water.
5	Mount clamps	Bottom pipe around 3 ft height; next at 4, 5, 6 ft.
6	Set slope	3-4 degree slope, about 3.5-5 inch drop over 6 ft.
7	Connect pump	300-500 LPH pump from 20L black reservoir.
8	Test with plain water	Run 2 hours before adding plants; fix leaks first.

Important: organic NFT is more difficult than grow bags. Start NFT with plain water test and simple leafy greens. Keep Jeevamrutham highly filtered. Biofilm and clogging are the main enemies.

8. Phase 4: Zone C Water, IoT And Fertilizer Hub

Back side should become the service area, but do not overload one small point.

Item	Placement	Safety rule
500L tank	Back/side near strong wall or beam	Needs strong stand; full tank weighs 500+ kg.
20L NFT reservoir	Below NFT pipes on right side	Must sit in tray, covered from sunlight.
Filter unit	Between rainwater/tank and farm lines	Easy access for cleaning.
ESP32 box	Back wall, shaded, waterproof IP65	Never place on floor.
Battery/solar controller	Weatherproof box, ventilated	Keep away from water spray.
Fertilizer corner	Small raised tray area	No wet drum directly on tiles.

Water flow plan

- Rain gutter collects roof/terrace water into first-flush diverter.
- First dirty water is discarded, clean water enters settling/filter unit.
- Filtered water goes to 500L tank.
- Tank gravity-feeds drip lines for grow bags.
- NFT reservoir runs separately with pump and returns water by gravity.
- Overflow from any tray must go to terrace drain, not random floor flow.

9. Phase 5: Sensor And Camera Setup

Sensors should first observe. Do not make automation control everything until readings are stable.

Sensor and camera placement

Camera 1
Back-left high angle
sees grow bags

ESP32 Box
Back wall, shaded,
waterproof IP65

Camera 2
Back-right high angle
sees NFT pipes

Moisture sensors
1 each in tomato,
chilli, bhindi groups

pH + TDS + water temp
inside NFT reservoir

DHT22 + LDR
center shade,
not direct sun

Sensor and camera placement plan.

Device	Where to place	Beginner purpose
Moisture sensor A	One tomato/cucumber bag	High-water group reading.
Moisture sensor B	One chilli/brinjal bag	Medium-water group reading.
Moisture sensor C	One bhindi/beans bag	Low-water group reading.
pH sensor	NFT reservoir	Prevent root nutrient problems.
TDS sensor	NFT reservoir	Watch solution strength.
DS18B20	NFT reservoir water	Heat warning.
DHT22	Center shade at 3 ft height	Air temp/humidity.
ESP32-CAM 1	Back-left high angle	Grow bag disease observation.
ESP32-CAM 2	Back-right high angle	NFT and right wall observation.

10. Soil Mix And Planting From Zero

Do not fill bags with random mud. Container soil must drain, breathe and feed microbes.

Ingredient	Ratio	Purpose
Cocopeat	40%	Moisture holding and lightness.
Forest sand / coarse sand	20%	Drainage and weight balance.
Vermicompost / mature compost	30%	Nutrition and biology.
Charged biochar / charcoal	10%	Water, nutrients and microbial housing.
Neem cake	1 handful per medium bag	Root pest protection.

Filling steps

- Place bag on shelf/tray, not on floor.
- Add a few broken tile pieces or coco chips at bottom only if drainage holes are large.
- Mix soil outside the bag first so it is even.
- Fill 80-85%, not to the top. Leave watering space.
- Water once and wait one day before transplanting.
- After planting, mulch the top with dry leaves to reduce heat.

11. First 90 Days Planting Plan

This plan avoids overload. You learn with strong crops first, then add difficult crops.

Period	Do this	Do not do this
Week 1	Build protection, shelf, NFT frame, water test	Do not buy all seedlings yet.
Week 2	Plant chilli, bhindi, brinjal, herbs in grow bags	Do not overwater daily by habit.
Week 3	Start tomato and cucumber with stakes/trellis	Do not leave vines unsupported.
Week 4	Start NFT with mint/methi first	Do not add thick organic liquids to NFT.
Month 2	Add coriander/palak if temperature allows	Do not ignore summer shade.
Month 3	Expand to full bag count if drainage is clean	Do not scale if leaks/smell/insects are unresolved.

12. Tools And Material List

Buy in layers. This prevents wasting money.

Category	Items	Approx cost range
Terrace protection	Trays, pallets/stands, drain mesh, sealant	Rs. 2,000-6,000
Grow bags	15x15, 24x24, 12x12 bags	Rs. 1,500-3,000
Shelf	GI/bamboo frame, planks/mesh	Rs. 1,500-5,000
NFT	PVC pipes, caps, pump, tubes, clamps, net pots	Rs. 2,000-4,500
Water	Tank, filter, drip pipes, valves	Rs. 3,000-8,000
IoT	ESP32, sensors, relay, wires, box	Rs. 4,000-8,000
Organic inputs	Cocopeat, compost, neem cake, jaggery	Rs. 1,500-4,000
Safety	MCB/RCCB, waterproof boxes, cable ducts	Rs. 2,000-5,000

13. Common Problems I Would Expect On Your Terrace

This is where experience matters. These problems are normal; the solution is preparation.

Problem	Why it happens	Prevention
Water leaking below	Standing water, tile joints, no trays	Raised system + drainage + weekly inspection.
NFT water too hot	Reservoir in sun, Chittoor summer	Shade reservoir, white cover, airflow, water temp sensor.
Plants dry fast	Tile heat and wind	Mulch, morning watering, shade net.
Fungus in monsoon	Hill humidity + wet leaves	Morning watering only, airflow, neem/buttermilk preventive.
Pests from existing plants	Old plant line can host insects	Inspect underside weekly, prune, neem spray.
Center becomes messy	Pots slowly occupy walkway	Walkway is sacred: keep empty always.
Electrical risk	Water + loose wires	Elevated wires, waterproof boxes, MCB/RCCB.

14. Daily, Weekly, Monthly Routine

This routine is more important than expensive technology.

Frequency	Checklist
Daily morning	Check water level, pump sound, leaf wilting, soil moisture, NFT flow, any floor water.
Daily evening	Look for pests, tie vines, check camera view, close fertilizer containers.
Weekly	Clean drain mesh, inspect leaf undersides, apply Jeevamrutham, check shade-net ropes.
Every 15 days	Calibrate pH sensor if installed, prune dead leaves, check pipe joints.
Monthly	Flush NFT, inspect waterproofing, clean trays, recharge compost, review records.
Before rain	Cover fertilizer drums, clear drains, secure trellis, check overflow pipes.

15. Beginner Rules That Save The Farm

If you remember only one page, remember this.

- No grow bag directly on terrace floor.
- No water tank without proper stand and weight planning.
- No electrical item on floor.
- No raw kitchen waste in grow bags.
- No spraying in afternoon heat.
- No full automation until manual system works.
- No overcrowding plants in first 90 days.
- No blocked center walkway.
- No ignoring first disease sign.
- No trusting memory: write dates, batches and plant response.

The mindset: your terrace is not just a garden. It is a small agricultural engineering system. Water, heat, roots, weight, microbes and electricity must all be controlled.

16. Printable Build Checklist

Use this as your actual work checklist.

Done	Task
■	Measure terrace exact length, width and drain direction.
■	Mark center walkway with chalk.
■	Repair cracks and parapet joints.
■	Install raised base/trays under future grow zones.
■	Build left-side grow bag shelf.
■	Mount right-side NFT clamps without pipes first.
■	Install water tank/stand and reservoir tray.
■	Test NFT pipes with plain water for 2 hours.
■	Fill first 10-15 grow bags only.
■	Plant first beginner crops.
■	Install sensors after plants and water system are stable.
■	Start daily farm notebook.

17. Final Recommended First Build

This is my practical build recommendation for your exact terrace.

Build item	Recommended first version
Grow bags	Start with 12-15 bags, not 30. Expand after 45 days.
NFT	Install all 4 pipes physically, but run only 2 pipes first.
Water tank	Use 200-300L first if structural doubt exists; upgrade to 500L after confirming stand and leakage.
Shade net	Install hooks now; use 50% shade net in summer.
Fertilizer	Use compost + Jeevamrutham first; add FPJ after flowering.
Automation	Start with monitoring alerts only; automation after 1 month of reliable readings.
Marketing	Do not sell first harvest. Use it to learn quality, yield and timing.

My honest advice: your terrace can support the project, but success will come from clean layout, drainage discipline, and gradual plant loading. Build the infrastructure in one phase, but activate crops and automation step by step.